

Soil Resources of the Indian Sub-continent

1. Soil forms the upper layer of the earth's crust capable of supporting life.
2. It is made up of loose rock materials and humus.
3. The soil forming processes are mainly influenced by the parent rock, climate, vegetation and animal life.

Importance of Soil Resources

1. Soil is an extremely important resource, especially in agricultural countries like India, Pakistan and Bangladesh.
2. Most food items, like rice, wheat, pulses, fruits and vegetables and much of our clothing are derived from the soil directly or indirectly.
3. Soil also gives us firewood, timber, rubber, fibers, etc. Food like milk, meat and eggs are obtained indirectly from the soil. Flowers, grass, plants and trees are also grown out of soil.

Soil Erosion and its types

1. Removal of top layer of soil when it is exposed to wind and rain, is easily blown or washed away. This condition is known as soil erosion.
2. Basically, soil cover is removed by two powerful agents — (i) Running water, (ii) Wind.

Types of Soil found in India

1. Indian Council of Agricultural Research (ICAR) divides Indian soils into eight groups : (a) Alluvial soil (b) Black soil (c) Red soil (d) Laterites and Lateritic soil (e) Arid and Desert soil (f) Saline and Alkaline soil (g) Forest soil (h) Peaty and other organic soil. However, Indian soils are generally divided into four broad types : (1) Alluvial soils (2) Regur soils (3) Red soils and (4) Laterite soils.

Alluvial Soils

1. This is the most important and widespread category. It covers 40% of the land area. In fact the entire Northern Plains are made up of these soils.

2. They have been brought down and deposited by three great Himalayan rivers- Sutlej, Ganga and Brahmaputra and their tributaries.
3. Through a narrow corridor in Rajasthan they extend to the plains of Gujarat.
4. They are common in Eastern coastal plains and in the deltas of Mahanadi, Godavari, Krishna and Cauveri.
5. Crops Grown : Suitable for Kharif & Rabi Crops like cereals, Cottons, Oilseeds and sugarcane. The lower Ganga-Brahmaputra Valley is useful for jute cultivation.

Regur or Black Soils

1. These soils are of volcanic origin. These soils are black in colour and are also known as black soils.
2. Since, they are ideal for growing cotton, they are also called black cotton soils, in addition to their normal nomenclature of Regur soils.
3. These soils are most typical of the Deccan trap (Basalt) region spread over north-west Deccan plateau and are made up of lava flows.
4. They cover the plateaus of Maharashtra, Saurashtra, Malwa and southern Madhya Pradesh and extend eastward in the south along the Godavari and Krishna Valleys.
5. Crops Grown : Cotton, Jowar, Wheat, Sugarcane, Linseed, Gram, Fruit & Vegetable.

Red Soils

1. Formed by weathering of crystalline and metamorphic mixture of clay and sand.
2. These soils are developed on old crystalline Igneous rocks' under moderate to heavy rainfall conditions.
3. They are red in colour because of their high Iron-oxide (FeO) content.
4. They are deficient in phosphoric acid, organic matter and nitrogenous material.
5. Red soils cover the eastern part of the peninsular region comprising Chhotanagpur plateau, Odisha (Orissa), eastern Chhattisgarh, Telangana, the Nilgiris and Tamil Nadu plateau.
6. They extend northwards in the west along the Konkan coast of Maharashtra.

7. Crops Grown : Wheat, Rice, Millets, Pulses.

Laterite Soils

1. The Laterite soils are formed due to weathering of lateritic rocks in high temperatures and heavy rainfall with alternate dry and wet period.
2. They are found along the edge of plateau in the east covering small parts of Tamil Nadu, Odisha and a small part of Chhotanagpur in the north and Meghalaya in the north-east.
3. Laterite soils are red in colour with a high content of iron-oxides; poor in Nitrogen and Lime.
4. Crops Grown : Unsuitable for agriculture due to high content of acidity and inability to retain moisture.

Arid & Desert Soil

1. Region : NW India. Covers entire area of the west Aravalis in Rajasthan and parts of Haryana, Punjab & Gujarat.
2. Characteristics : Rich in Phosphates and Calcium but deficient in Nitrogen and humus.
3. Corps Grown : Fertile if irrigated e.g. Ganga Nagar area of Rajasthan (Wheat basket of Rajasthan).